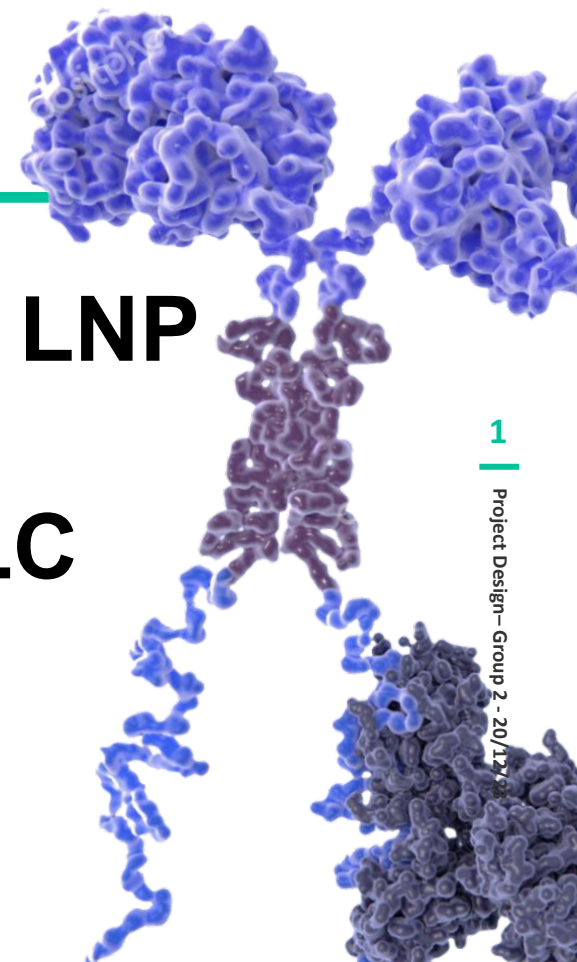


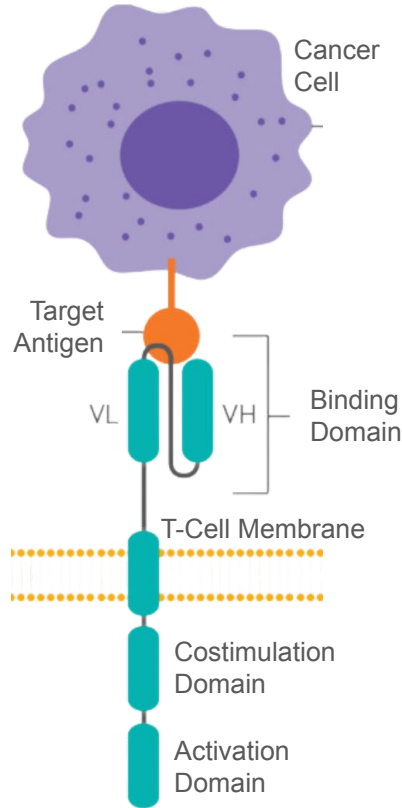
Inhalation of saRNA loaded LNP for in vivo induction of CAR-macrophages in NSCLC

Presented by Nouchine Bouchiat, Leila Diouri and Kilian Zell



CAR-T to Treat Cancer

- CAR: **engineered receptor** that enhances **recognition of cancer** cell surface antigen
- **CAR-T** cell therapy was shown to be an **efficient** treatment for **hematological malignancies**



Limitations

Average Cost per Patient

\$ 800k



<9%

Response Rate
in Solid Tumors



Life Danger

Life threatening side
effects such CRS

Neelapu SS, Locke FL, Bartlett NL, et al. Axicabtagene Ciloleucel CAR T-Cell Therapy in Refractory Large B-Cell Lymphoma. *N Engl J Med*. 2017;377(26):2531-2544. doi:10.1056/NEJMoa1707447
Sharma P, Hu-Lieskovan S, Wargo JA, Ribas A. Primary, Adaptive, and Acquired Resistance to Cancer Immunotherapy. *Cell*. 2017;168(4):707-723. doi:10.1016/j.cell.2017.01.017
<https://biocannx.com/recherche/made-canada-car-t-platform?lang=fr>
Maakaron, J. E., Hu, M., & El Jurdi, N. (2022). Chimeric antigen receptor T cell therapy for cancer: Clinical applications and practical considerations. *BMJ*, 378, e068956. <https://doi.org/10.1136/bmj-2021-068956>
<https://www.primetherapeutics.com/news/real-world-car-t-treatment-costs-can-range-from-700000-to-1-million-2/>
Hou, B., Tang, Y., Li, W., Zeng, Q., & Chang, D. (2019). Efficiency of CAR-T Therapy for Treatment of Solid Tumor in Clinical Trials: A Meta-Analysis. *Journal Name*, Volume 2019, Article ID 3425291. <https://doi.org/10.1155/2019/3425291>
Shah NN, Fry TJ. Mechanisms of resistance to CAR T cell therapy. *Nat Rev Clin Oncol*. 2019;16(6):372-385. doi:10.1038/s41571-019-0184-6

CAR-M Therapy

- **Macrophages** have the highest infiltration rate in **solid tumor**
- Induction of **CAR-M** for specialized **targeting**, **engulfment** and **cytotoxic activation**

Phase I Clinical Trials

①



carisma
THERAPEUTICS

Chimeric adenoviral vector delivers the CAR transgene and induces an M1 phenotype

Injection of millions of engineered macrophages, considering the difficulty of in vitro expansion of macrophages the application is challenging.

②



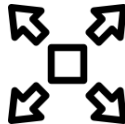
CAR mRNA transfected directly into PBMCs

Transient expression without the necessity of viral vectors or cell amplification Transfection with MaxCyte GT System™ (FDA-approved electroporation system)



In Vitro
Induction

+



In Vitro
Expansion

=

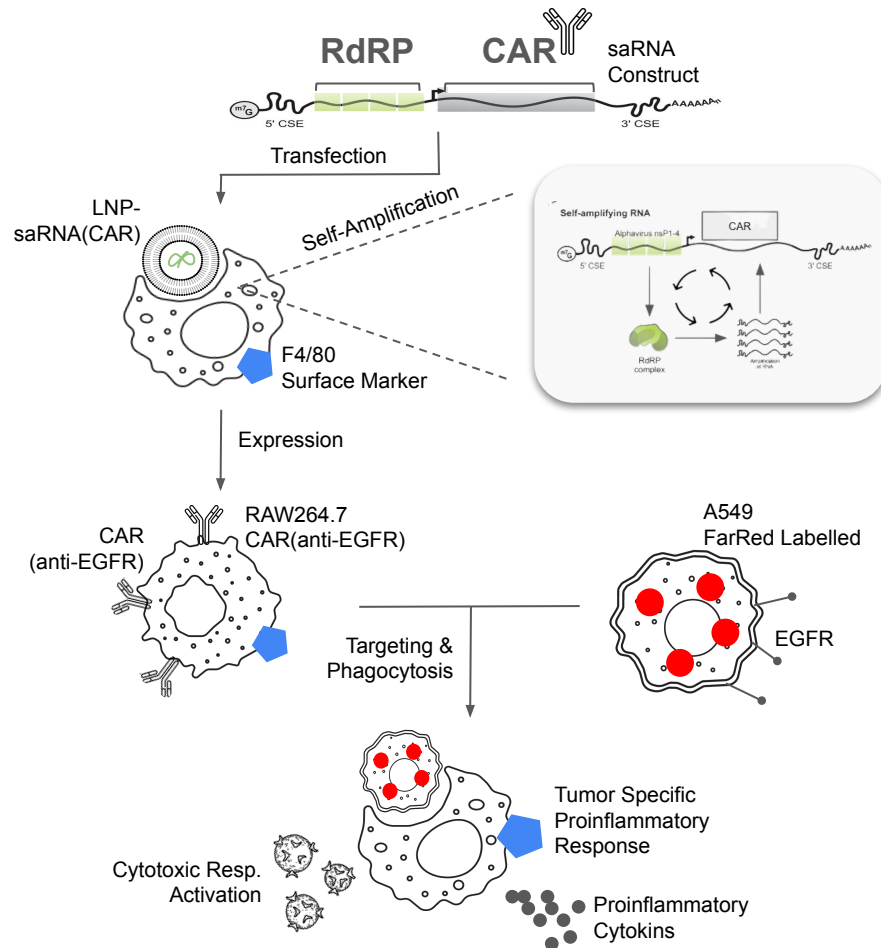


Not Cost
Effective

Proposed Solution:

In-Vivo Induction

- Use **self-amplifying RNA (saRNA)** instead of mRNA to **improve the expression** of CAR in Macrophages
- **Deliver** the construct in a **Lipid Nanoparticle (LNP)** of **optimized composition**



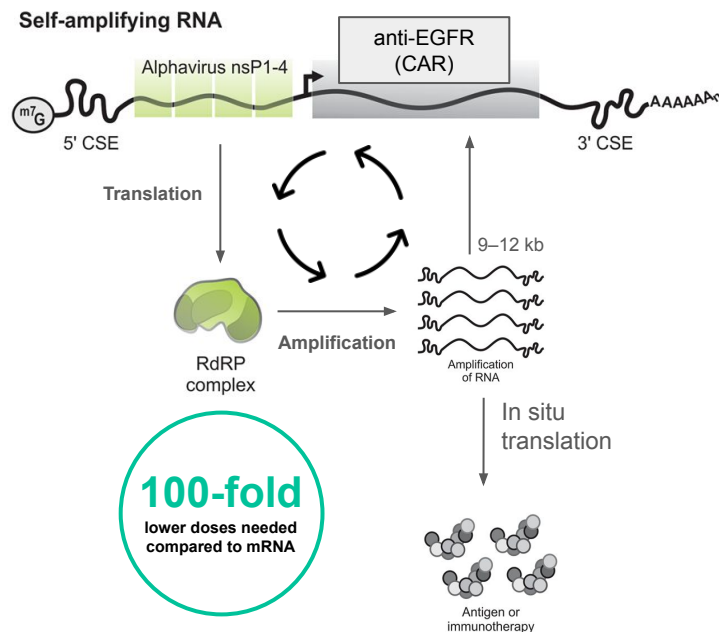
Unique Value Proposition

Our Solution: saRNA for In-Vivo Induction ...

- Long coding region with RDRP and target protein
- Upon delivery, the mRNA serves as a template for RDRP production
- Also functions as a genomic template for RDRP-mediated replication

→ **Allows efficient production from limited endosomal RNA escape**

RDRP: RNA-dependent RNA polymerase

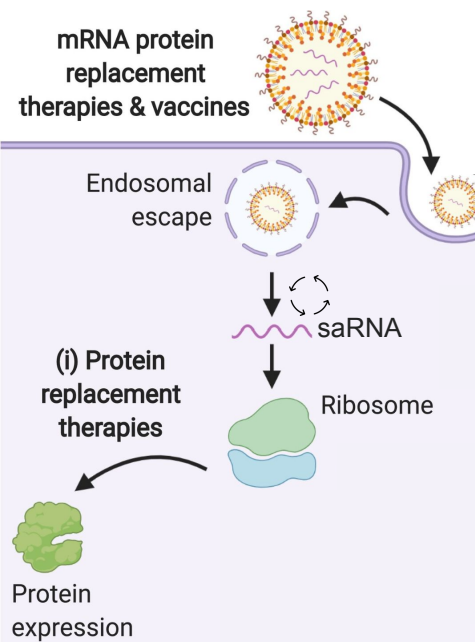
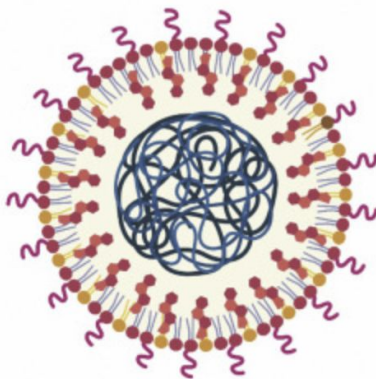


Our Solution: ... Delivered in Lipid Nanoparticles



Rational

Ye et al. (2022) screened LNPs for mRNA-CAR delivery in macrophage-like cells
(in vitro)



Swingle, K. L., Hamilton, A. G., & Mitchell, M. J. (2021). Lipid Nanoparticle-Mediated Delivery of mRNA Therapeutics and Vaccines. *Molecular Medicine*, 27(6), 616-617.

<https://doi.org/10.1016/j.molmed.2021.03.003>

Ly, H. H., Daniel, S., Soriano, S. K. V., Kis, Z., & Blakney, A. K. (2022). Optimization of Lipid Nanoparticles for saRNA Expression and Cellular Activation Using a Design-of-Experiment Approach. *Molecular pharmaceutics*, 19(6), 1892-1905. <https://doi.org/10.1021/acs.molpharmaceut.2c00032>

Blakney, A. K., McKay, P. F., Hu, K., Samnuan, K., Jain, N., Brown, A., Thomas, A., Rogers, P., Polra, K., Sallah, H., Yeow, J., Zhu, Y., Stevens, M. M., Geall, A., & Shattock, R. J. (2021). Polymeric and lipid nanoparticles for delivery of self-amplifying RNA vaccines. *Journal of controlled release : official journal of the Controlled Release Society*, 338, 201-210.

<https://doi.org/10.1016/j.jconrel.2021.08.029>

Model Candidates



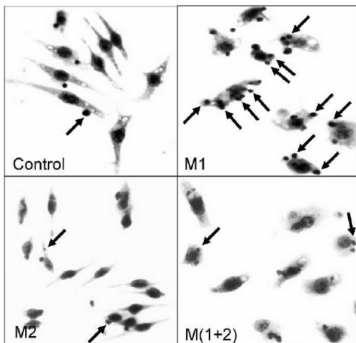
RAW264.7: Cell Model for Macrophages

RAW264.7

Murine macrophage cell line commonly used to study macrophage biology



In Vitro PoC



RAW264.7 co-cultured with apoptotic Jurkat cells. Attached and engulfed apoptotic cells are marked with arrows.

M1 macrophages are more efficient interacting with apoptotic cells than are M2 macrophages.

- Monocyte/macrophage-like cell lineage, originating from leukemia virus transformed cells derived from mice
- Capable of performing phagocytosis and mimicking certain aspects of primary macrophages
- Robust inflammatory response when challenged by LPS

LPS: Lipopolysaccharide

A549: Cell Model for NSCLC

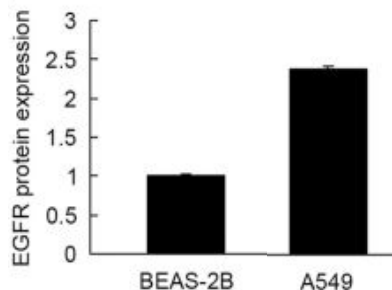
A549

Human adenocarcinoma cell line that represents a common subtype of NSCLC.



In Vitro PoC

- Adenocarcinomic human alveolar basal epithelial cells
- Responsible for the diffusion of substances across alveoli

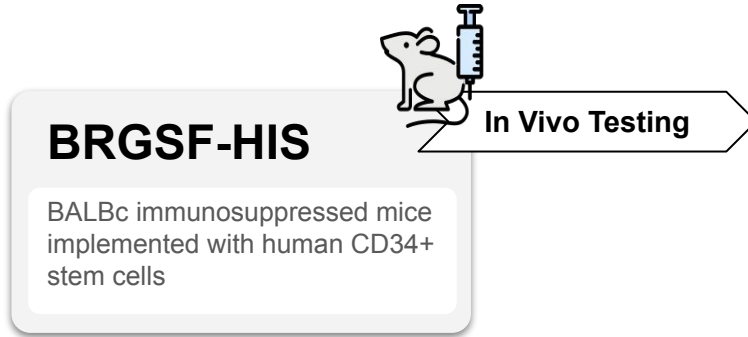


Expression of EGFR protein in A549 cells. EGFR protein expression, as assessed using western blot analysis.

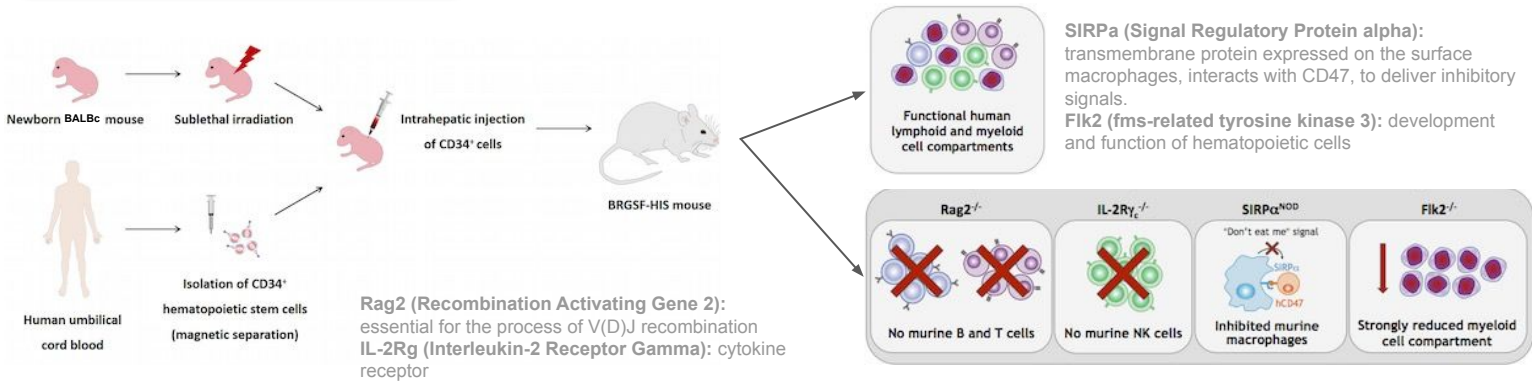
Driver Mutation: KRAS p.Gly12Ser (Hom)

BEAS-2B: normal human bronchial epithelial cell line
EGFR: Epidermal Growth Factor Receptor

BRGSF-HIS: Humanized Mouse Model



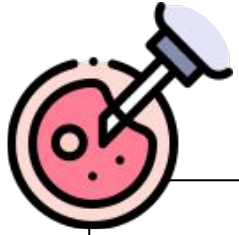
- Commercially available (*genoway*), possess all of the major human hematopoietic cell subsets, including T-cells and macrophages.



Experimental Methods



Two Step Experimental Workflow



In Vitro PoC



In Vivo Testing

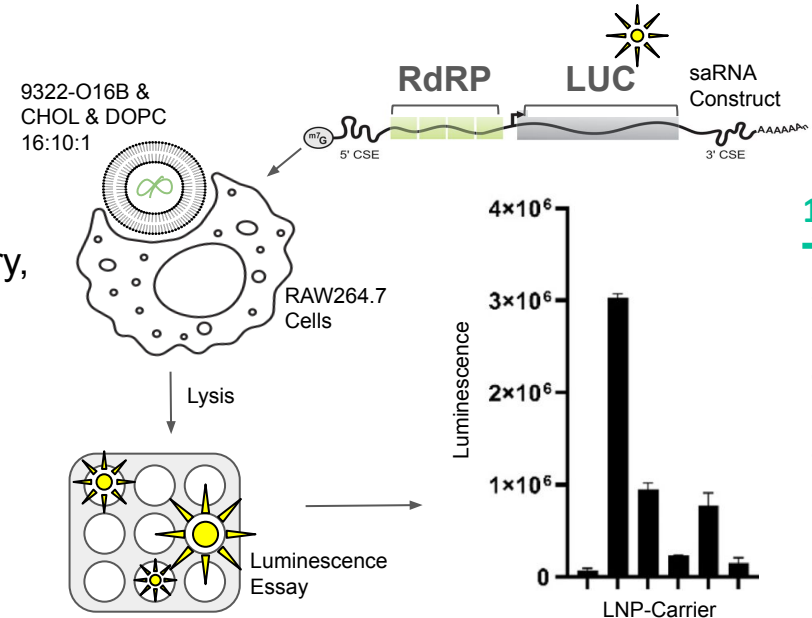
Purpose

- Investigate the **fundamental mechanisms** and interactions involved
 - Establish the **feasibility**, **optimize** conditions, and anticipate **challenges**
- Assess the safety, **effectiveness**, and overall **performance** of the proposed intervention
 - **Paving the way** for further **clinical** development and potential **therapeutic use**.

LUC-Essay for LNP-Delivery Assessment



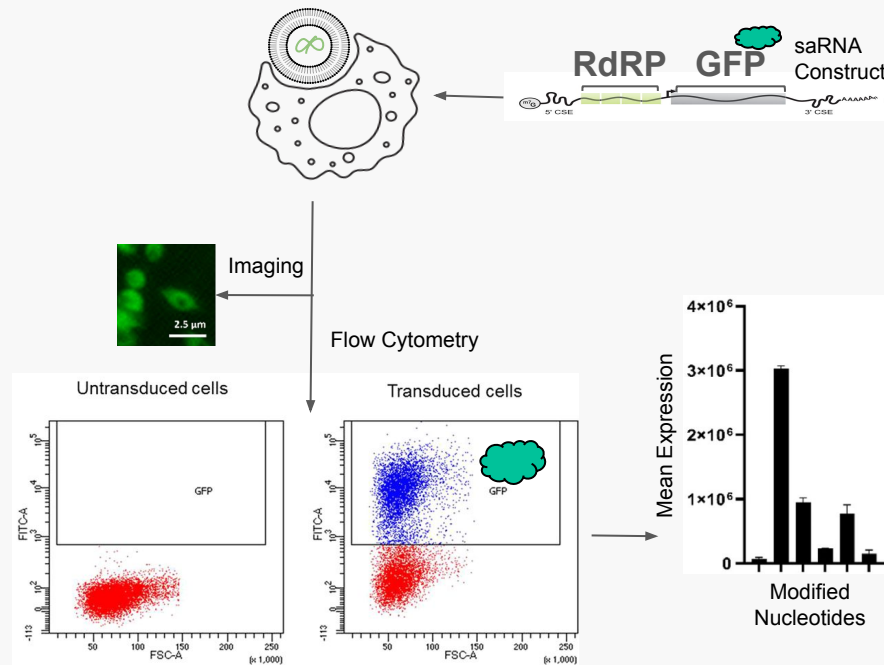
- Ye et al. (2022) screened LNPs for mRNA delivery in macrophage-like cells
- To validate their formulation for saRNA delivery, we will assess transfection efficiency using a luciferase assay.
- RAW264.7 cells transfected with luciferase-expressing saRNA encapsulated in the LNP of interest.



GFP Reporter Assay of saRNA Modifications to Optimize Delivery

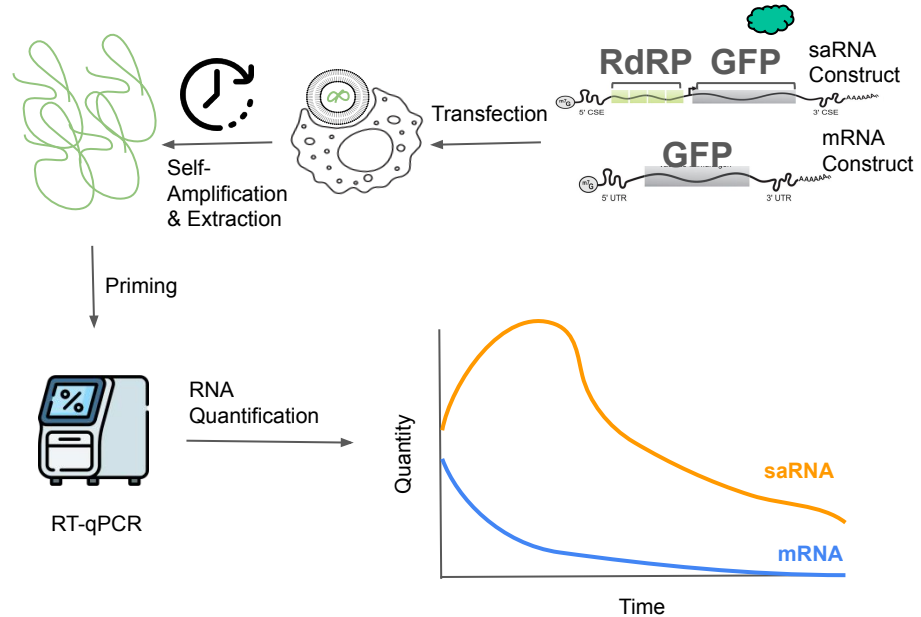
NOT
PRESENTED

- Differently modified nucleotides and their combinations will be incorporated into GFP-expressing saRNA
- GFP expression of saRNA in RAW264.7 will be evaluated
- Trade-off between expression rate and the mean expression



RT-qPCR to Evaluate saRNA Self-Amplification

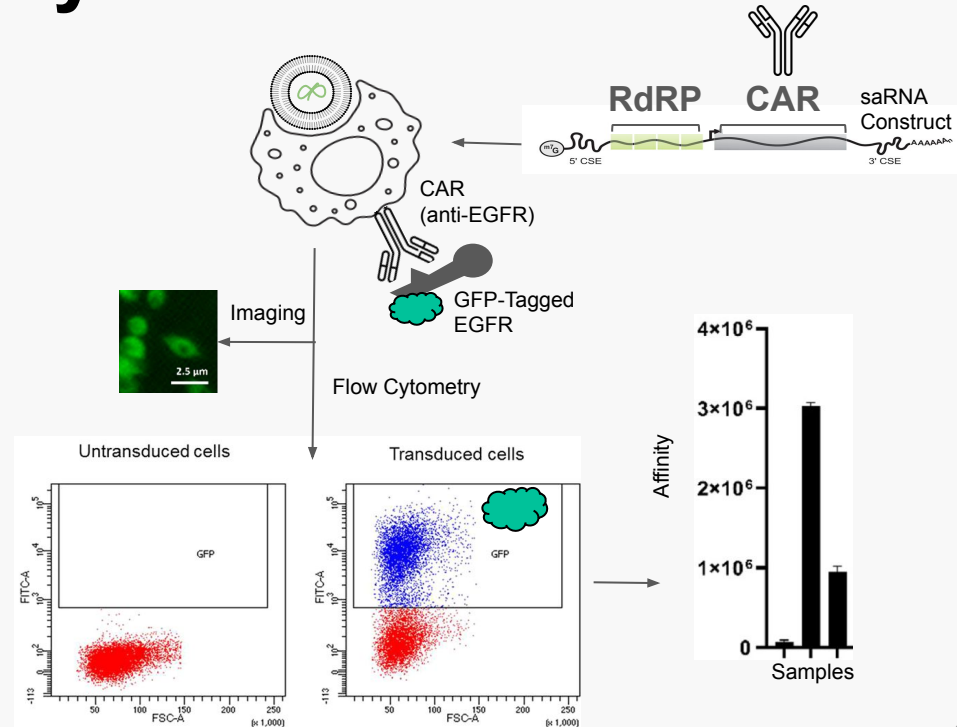
- Compare mRNA vs saRNA delivery kinetics of GFP sequence
- Asses the self-amplification capacity of saRNA
- Followed by similar experiment on GFP expression using Western Blot



GFP-Tagged EGFR to Assess CAR-EGFR Affinity

NOT
PRESENTED

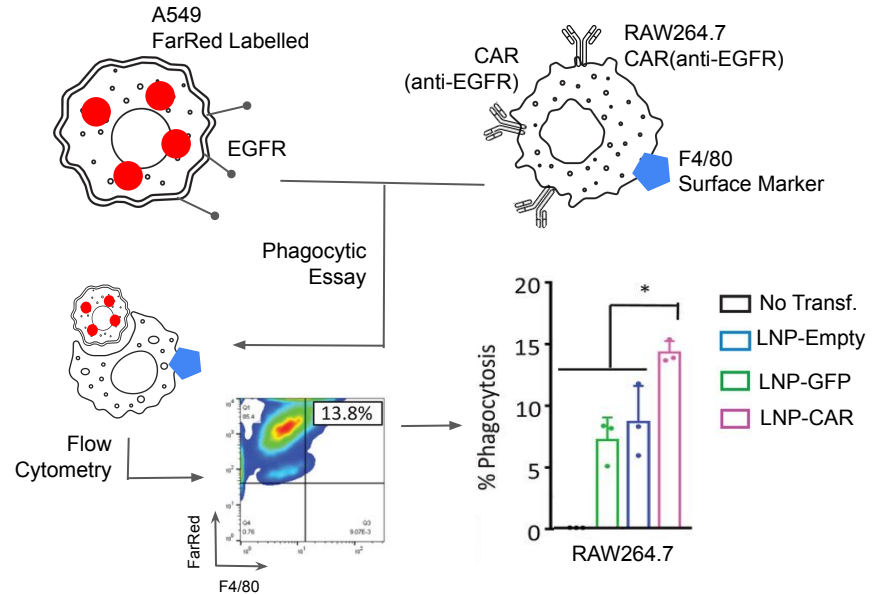
- Investigate the success of CAR M induction following the treatment
- Use GFP-tagged soluble EGFR to bind induced CAR(anti-EGFR)
- Followed by flow cytometry to quantify CAR expression



Carter, R. E., & Sorkin, A. (1998). Endocytosis of functional epidermal growth factor receptor-green fluorescent protein chimera. The Journal of biological chemistry, 273(52), 35000–35007. <https://doi.org/10.1074/jbc.273.52.35000>

Phagocytosis Assay to Assess CAR(anti-EGFR) Efficiency In-Vitro

- LPS used as adjuvant to promote pro-inflammatory phenotype
- Far Red-stained A549 are engulfed by RAW264.7 treated with LNP-saRNA(anti-EGFR)
- Flow cytometry (FarRed vs F4/80) to quantify the phagocytic rate

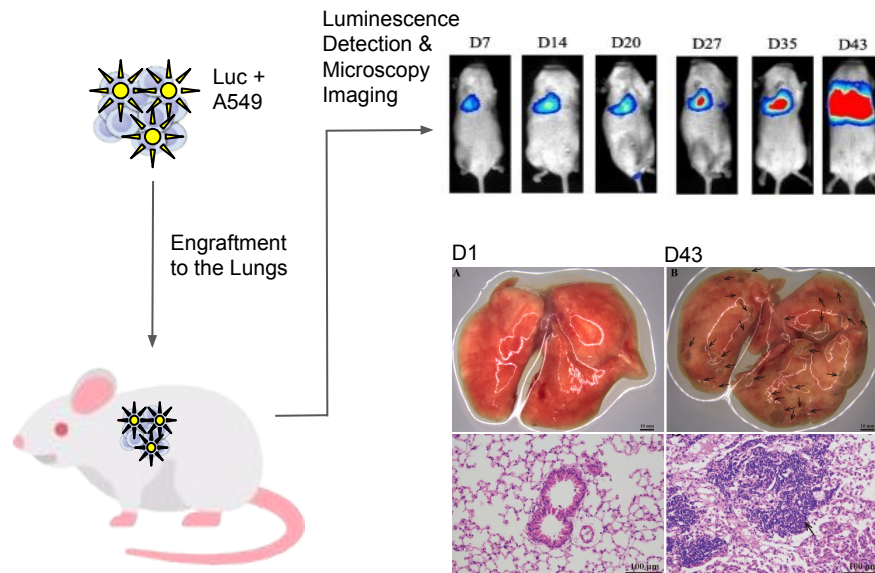


FarRed: labeling agent in biological imaging and fluorescence-based assays
 F4/80: well-known marker for murine macrophages, often used to identify RAW264.7

Orthotopic Xenograft for In-Vivo Testing in BRGSF-HIS



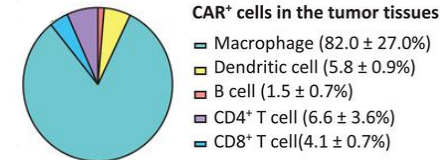
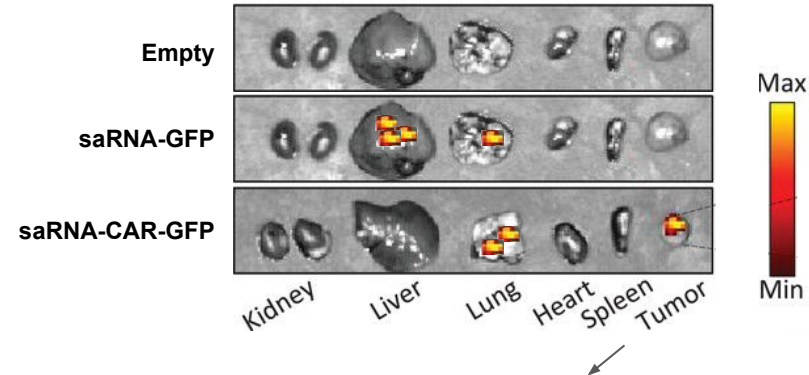
- Allow the evaluation of tumor behavior in an organ-specific microenvironment
- Transfected NSCLC expressing LUC reporter gene
→ **Allows monitoring of tumor progress by in vivo imaging**



Ex-Vivo Fluorescence to Localize Protein Expression

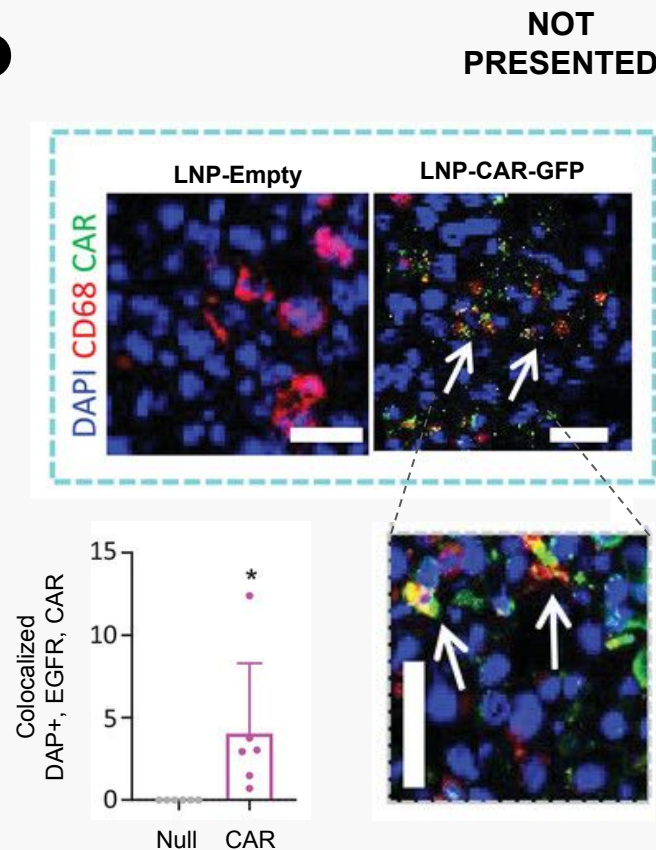


- Inhalation of LNP loaded with bicistronic replicons:
LNP(saRNA-antiEGFR-GFP)*
- Measure fluorescence in major organs and tumor tissues after 48h
- Flow cytometry to determine the distribution of CAR⁺ cells in the tumor tissues



Fluorescence Images to Quantify Colocalization

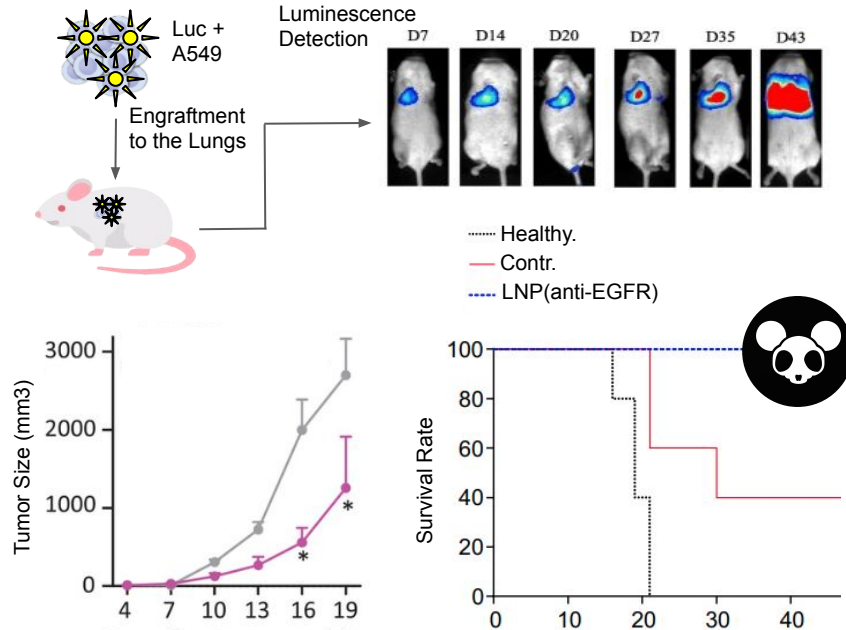
- Inhalation of LNP loaded with bicistronic replicons:
LNP(saRNA-antiEGFR-GFP)*
- mCherry-EGF complex to visualize tumor cells
- Fluorescence images and quantification of A549 cancer cell (red) and CAR (green)-expressing macrophage colocalization





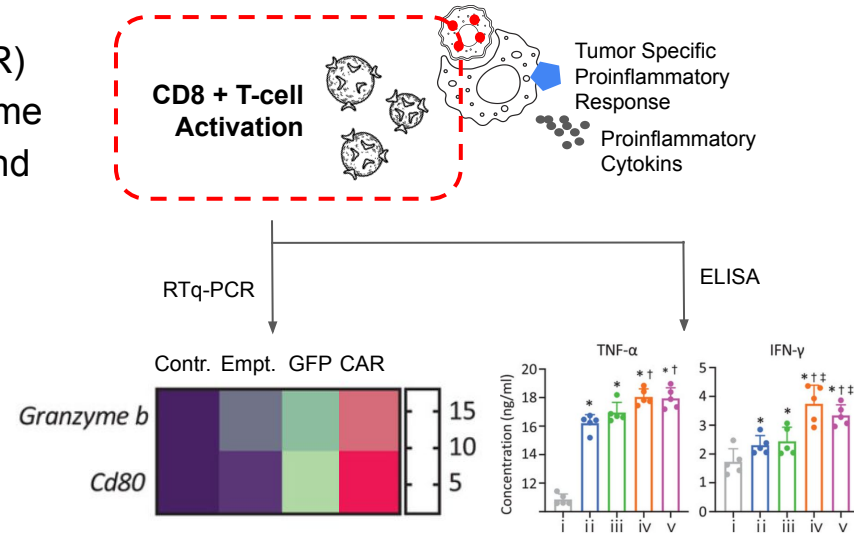
LUC Reporter to Monitor Tumor Regression After Treatment

- Transfected NSCLC expressing LUC reporter gene engrafted in humanized mouse (BRGSF-HIS)
- Luminescence detection to evaluate tumor size after treatment
- Evaluate the survival rate over time



qRT-PCR and ELISA to Evaluate CD8+ T-Cell Activation

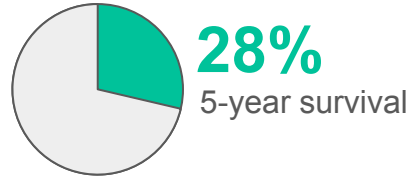
- qRT-PCR after inhalation of LNP(CAR) to evaluate the expression of Granzyme b (proxy for CD8+ T-cell activation) and CD80 (APC co-stimulatory signal)
- ELISA to assess the level of proinflammatory cytokines secreted after presumed activation



Granzyme b: protease released during cytotoxic response
CD80: co-stimulatory expressed on the surface of aAPCs, interacts with CD28 of T cells

Developed to Cure NSCLC

- Leading cause of cancer-related deaths globally
- EGFR is overexpressed and provides a distinct and targetable surface antigen for CAR therapy

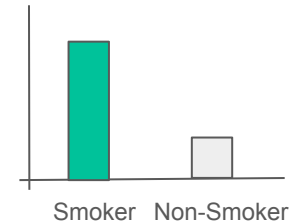


Rank in cancer death

n°1

People diagnosed every year

2 Million



Any questions or feedbacks ?

